Stage 1 Hard Drill — With Solutions

Chapter 6.2 — Optimal Stopping Theory: the Secretary Problem

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — Twelve Candidates, Exact Cut-off

Twelve candidates arrive in uniformly random order. You see only relative ranks, must decide on the spot (no recall) and win only by hiring the overall best. Find the optimal number s to skip, the exact winning probability, and $\mathbb{P}$(nobody hired).

Solution. Hire a record at position j iff $\sum_{i=j}^{n-1} \frac{1}{i} \le 1$:

$$\sum_{i=5}^{11} \tfrac{1}{i} = 0.9365 \le 1, \qquad \sum_{i=4}^{11} \tfrac{1}{i} = 1.1865 > 1$$

So the first acceptable record is $j = 5$: skip $\boxed{s = 4}$ (and $n/e = 4.41$ agrees). Then

$$\mathbb{P} = \frac{4}{12} \sum_{i=4}^{11} \frac{1}{i} = \frac{1}{3}(1.1865) = \boxed{\tfrac{32891}{83160} \approx 0.3955}$$

Neighbours: $s = 3$ gives 0.3800, $s = 5$ gives 0.3902. Nobody is hired iff the best is among the first s:

$$\mathbb{P}(\text{no hire}) = \tfrac{s}{n} = \tfrac{1}{3}$$

⋆ *Intuition:* the cut-off is the point where the *expected number of future records* drops to one; before that, waiting is worth more than grabbing.

Q2 — When Will You Stop? (Law of the Stopping Time)

Ten candidates, skip the first 3, then hire the first record. Find the law of the hiring position T, then $\mathbb{P}$(win) and $\mathbb{E}$[#interviews].

Solution. You stop at $j > 3$ iff j is a record (probability $\frac{1}{j}$) *and* the best of the first $j-1$ is in the first 3 (probability $\frac{3}{j-1}$):

$$\mathbb{P}(T = j) = \frac{3}{j(j-1)} : \quad \tfrac{1}{4},\ \tfrac{3}{20},\ \tfrac{1}{10},\ \tfrac{1}{14},\ \tfrac{3}{56},\ \tfrac{1}{24},\ \tfrac{1}{30}$$

for $j = 4, \dots, 10$. They sum to $\frac{7}{10}$; the remaining $\frac{3}{10} = \frac{s}{n}$ is the no-hire event. A record at j is the overall best with probability $\frac{j}{n}$, so

$$\mathbb{P}(\text{win}) = \sum_{j=4}^{10} \mathbb{P}(T = j) \cdot \frac{j}{10} = \frac{3}{10} \sum_{i=3}^{9} \frac{1}{i} = \boxed{\tfrac{3349}{8400} \approx 0.3987}$$

$$\mathbb{E}[T] = \sum_{j=4}^{10} j \cdot \frac{3}{j(j-1)} + 10 \cdot \tfrac{3}{10} = 3(1.3290) + 3 = \boxed{6.987}$$

Check: $\mathbb{E}[T] = n\,\mathbb{P} + s = 3.987 + 3$.

▶ *Speed trick:* the factor j in $\mathbb{E}[T]$ cancels the $\frac{1}{j}$ in the law, leaving a pure harmonic sum.

Q3 — How Forgiving Is the Cut-off? ($n = 100$)

A manager facing $n = 100$ candidates skips exactly half, $s = 50$. How much win probability does that cost compared with the optimum? Which cut-offs s achieve at least 0.36, and at least 0.35?

Solution. $\mathbb{P}_{100}(s) = \frac{s}{100} \sum_{i=s}^{99} \frac{1}{i}$:

s	20	25	30	37	50	60
$\mathbb{P}$	.326	.350	.365	.371	.349	.309

Optimum $s = 37$ gives 0.3710; skipping half gives 0.3491:

$$\text{loss} = 0.3710 - 0.3491 = \boxed{0.022}$$

Scanning all s:

$$\mathbb{P} \ge 0.36 \iff 28 \le s \le 45, \qquad \mathbb{P} \ge 0.35 \iff 25 \le s \le 49$$

Note $s = 25$ (0.3504) and $s = 50$ (0.3491) nearly tie, echoing the exact asymptotic tie $\phi(\frac{1}{4}) = \phi(\frac{1}{2}) = \frac{\ln 2}{2}$.

⋆ *Intuition:* the success curve is a broad plateau — anywhere from about a quarter to a half of the pool as exploration is within 0.025 of optimal.

Q4 — Stop on the Last Six

A fair die is rolled 10 times, one roll at a time. After each roll you may stop; you win iff the roll you stop on is the *last* six of the ten. Find the optimal rule and winning probability. Repeat for "the last roll showing 5 or 6".

Solution. Rolls are independent with $p = \frac{1}{6}$, odds $r = \frac{p}{1-p} = \frac{1}{5}$. Sum odds from the back until reaching 1:

$$5 \cdot \tfrac{1}{5} = 1 \ \Rightarrow\ \text{start at roll 6}$$

$$\mathbb{P}(\text{win}) = \left(\tfrac{5}{6}\right)^5 \cdot 1 = \boxed{\tfrac{3125}{7776} \approx 0.402}$$

This equals $\mathbb{P}(\text{exactly one six in rolls 6–10}) = 5 \cdot \frac{1}{6}\left(\frac{5}{6}\right)^4$. Since the odds sum is *exactly* 1, starting at roll 5 ties.

For $\{5, 6\}$: $p = \frac{1}{3}$, $r = \frac{1}{2}$; two rolls give odds sum 1, so start at roll 9:

$$\mathbb{P}(\text{win}) = \left(\tfrac{2}{3}\right)^2 \cdot 1 = \boxed{\tfrac{4}{9}}$$

▶ *Speed trick:* when the odds sum lands exactly on 1, the window is simply "exactly one success in the last m trials".

Q5 — Odds Theorem With Uneven Signals

Ten independent spike signals occur on consecutive days with probabilities

$$p = \left(\tfrac{1}{20}, \tfrac{1}{10}, \tfrac{1}{5}, \tfrac{1}{4}, \tfrac{7}{20}, \tfrac{3}{10}, \tfrac{3}{20}, \tfrac{2}{5}, \tfrac{1}{5}, \tfrac{3}{10}\right)$$

You want to act on the *last* spike. Find the optimal start day and winning probability, and show that starting one day earlier or later is worse.

Solution. Odds $r_k = \frac{p_k}{1-p_k}$ for the last three days:

$$r_{10} = \tfrac{3}{7}, \quad r_9 = \tfrac{1}{4}, \quad r_8 = \tfrac{2}{3}$$

$$\tfrac{3}{7} = 0.429, \quad \tfrac{3}{7} + \tfrac{1}{4} = 0.679, \quad +\tfrac{2}{3} = \tfrac{113}{84} = 1.345 \ge 1$$

So start on day $\boxed{8}$. Product of $(1-p_k)$ for days 8, 9, 10 is $\frac{3}{5} \cdot \frac{4}{5} \cdot \frac{7}{10} = \frac{42}{125}$:

$$\mathbb{P}(\text{win}) = \frac{42}{125} \cdot \frac{113}{84} = \boxed{\tfrac{113}{250} = 0.452}$$

Start day 7: $0.85 \cdot 0.336 \cdot 1.5217 = 0.435$. Start day 9: $0.56 \cdot 0.6786 = 0.380$. Both lose (a simulation of the day-8 rule gives 0.453).

⋆ *Intuition:* you start monitoring as soon as the *total odds of the remaining signals* reach one — no need for any dynamic programming.

Q6 — Poisson Alerts: Stop on the Last One

Alerts arrive as a Poisson process of rate $\lambda = 4$ per hour during a 3-hour session. You may act on one alert and win iff it is the *last* of the session. Find the best rule and winning probability. Repeat for $\lambda = \frac{1}{2}$ per hour and a 1.5-hour session.

Solution. Rule: act on the first alert after time $T - u$. You win iff there is exactly one alert in the final window of length u:

$$\mathbb{P}(\text{win}) = \lambda u\, e^{-\lambda u}$$

Maximise: derivative $\lambda e^{-\lambda u}(1 - \lambda u) = 0 \Rightarrow u = \frac{1}{\lambda}$:

$$u = \tfrac{1}{4}\ \text{h} \ \Rightarrow\ \text{start at 2:45}, \qquad \mathbb{P} = \boxed{\tfrac{1}{e} \approx 0.368}$$

For $\lambda = \frac{1}{2}$, $T = 1.5$: here $\lambda T = 0.75 < 1$, so even the full window is shorter than $\frac{1}{\lambda}$. Start immediately ($u = T$):

$$\mathbb{P} = 0.75\, e^{-0.75} = \boxed{0.354}$$

▶ *Speed trick:* this is the odds theorem with infinitesimal steps: "odds sum" becomes the *expected number of arrivals left*. Start when it equals 1, and win with $\mathbb{P}$(exactly one) $= \frac{1}{e}$.

Q7 — You See the Scores: Three Uniform Draws

Three iid Uniform$(0,1)$ numbers are revealed one at a time. You must stop on one and win iff it is the *largest* of the three. Find the optimal thresholds and the exact winning probability (compare with the no-information value $\frac{1}{2}$).

Solution. Work backwards. *Draw 2* (best so far m): accepting a record y wins with probability y; continuing means taking the forced last draw, which wins with $1-y$. So accept iff $y > m$ and $y > \frac{1}{2}$.

Value of continuing after draw 1 with best-so-far $m \geq \frac{1}{2}$:

$$A(m) = \int_m^1 y\,dy + m(1-m) = \tfrac{1}{2} + m - \tfrac{3}{2}m^2$$

and for $m < \frac{1}{2}$: $A(m) = \frac{3}{4} - \frac{m^2}{2}$.

Draw 1: accepting x wins with x^2 (both later draws must be below it):

$$x^2 \geq \tfrac{1}{2} + x - \tfrac{3}{2}x^2 \iff 5x^2 - 2x - 1 \geq 0$$

$$x \;\geq\; t_1 = \frac{1+\sqrt{6}}{5} \approx \boxed{0.690}$$

The winning probability integrates the best action:

$$\int_0^{1/2} \left(\tfrac{3}{4} - \tfrac{x^2}{2}\right)dx + \int_{1/2}^{t_1} \left(\tfrac{1}{2} + x - \tfrac{3}{2}x^2\right)dx + \int_{t_1}^1 x^2\,dx$$

$$= \boxed{\tfrac{293}{600} + \tfrac{2\sqrt{6}}{25} \approx 0.6843}$$

(a 10^6-run simulation gives 0.6845).

⋆ *Intuition:* thresholds fall as draws run out $(0.69,\ 0.5)$, and seeing actual values lifts the win chance from 0.50 to 0.68.

Q8 — Settle for the Best or Second Best ($n = 4$)

Four candidates in random order, relative ranks only, no recall. Now you win if you hire the best *or* the second best. Find the optimal policy and the winning probability (compare with $\frac{11}{24}$ for "best only").

Solution. State (j, r): the j-th candidate has relative rank r among those seen. Probability that hiring now gives absolute rank ≤ 2:

$j=4:\ r{\leq}2 \to 1,$ else 0 $\qquad$ $j=3:\ r{=}1 \to 1,\ r{=}2 \to \frac{1}{2},\ r{=}3 \to 0$

$$j=2:\ r{=}1 \to \tfrac{5}{6},\ r{=}2 \to \tfrac{1}{6} \qquad j=1:\ \tfrac{1}{2}$$

Let C_j be the value of continuing optimally after candidate j:

$$C_3 = \tfrac{1}{4}(1+1+0+0) = \tfrac{1}{2}$$

$$C_2 = \tfrac{1}{3}\big(\max(1,\tfrac{1}{2}) + \max(\tfrac{1}{2},\tfrac{1}{2}) + \max(0,\tfrac{1}{2})\big) = \tfrac{2}{3}$$

$$C_1 = \tfrac{1}{2}\big(\max(\tfrac{5}{6},\tfrac{2}{3}) + \max(\tfrac{1}{6},\tfrac{2}{3})\big) = \tfrac{1}{2}\cdot\tfrac{3}{2} = \boxed{\tfrac{3}{4}}$$

Policy: skip candidate 1 (as $\frac{1}{2} < \frac{3}{4}$); hire candidate 2 iff best so far; hire candidate 3 iff best so far (second best so far is a tie); take whoever comes last if you get there.

⋆ *Intuition:* relaxing the goal from "the best" to "top two" lifts the success chance from 0.458 to 0.75.

Q9 — Minimise the Expected Rank ($n = 4$)

Four candidates with absolute ranks 1 (best) to 4 in random order; relative ranks only, no recall, you must hire someone. Find the policy minimising the *expected absolute rank* of the hire, and its value.

Solution. Given relative rank r among j seen, the expected absolute rank is $\frac{r(n+1)}{j+1} = \frac{5r}{j+1}$. Backward induction (smaller is better):

$$C_3 = \tfrac{1}{4}(1+2+3+4)\cdot 1 = \tfrac{5}{2} \quad \text{(forced last hire: mean rank } \tfrac{5}{2}\text{)}$$

Candidate 3: values $\frac{5}{4}, \frac{5}{2}, \frac{15}{4}$ against $C_3 = \frac{5}{2}$, so accept $r = 1$, tie at $r = 2$:

$$C_2 = \tfrac{1}{3}\left(\tfrac{5}{4} + \tfrac{5}{2} + \tfrac{5}{2}\right) = \tfrac{25}{12}$$

Candidate 2: values $\frac{5}{3}, \frac{10}{3}$ against $C_2 = \frac{25}{12}$, so accept only $r = 1$:

$$C_1 = \tfrac{1}{2}\left(\tfrac{5}{3} + \tfrac{25}{12}\right) = \boxed{\tfrac{15}{8} = 1.875}$$

Hiring the first candidate gives $\frac{5}{2}$. A brute-force check over all 24 orders and all relative-rank thresholds confirms $\frac{15}{8}$ (it is also what the secretary rule with $s = 1$ achieves at $n = 4$).

▶ *Speed trick:* the acceptance value $\frac{r(n+1)}{j+1}$ is linear in r, so every optimal policy is a threshold in relative rank. For $n = 10$ the optimum is $\frac{3223}{1260} \approx 2.56$, and the limit as $n \to \infty$ is about 3.87.

Q10 — Capstone: Thirty Candidates

For $n = 30$ candidates, find the optimal cut-off, $\mathbb{P}$(best), $\mathbb{P}$(nobody hired), $\mathbb{P}$(wrong hire), $\mathbb{P}$(best | hired), $\mathbb{E}$[#interviews], and the expected number of records after the cut-off. Then decide whether to hire records at $j = 11$ and $j = 12$.

Solution. Cut-off:

$$\sum_{i=12}^{29} \tfrac{1}{i} = 0.9418 \leq 1 < 1.0327 = \sum_{i=11}^{29} \tfrac{1}{i} \;\Rightarrow\; \boxed{s = 11}$$

$(n/e = 11.04)$. *Success:*

$$\mathbb{P} = \frac{11}{30}(1.0327) = \boxed{0.3787}$$

No hire: $\frac{11}{30} = 0.3667$. *Wrong hire:* $1 - 0.3787 - 0.3667 = 0.2547$. Given somebody is hired:

$$\mathbb{P}(\text{best} \mid \text{hired}) = \frac{0.3787}{19/30} = 0.598$$

Interviews: $\mathbb{E}[T] = n\,\mathbb{P} + s = 30(0.3787) + 11 = \boxed{22.36}$. *Records after the cut-off:*

$$\sum_{j=12}^{30} \frac{1}{j} = H_{30} - H_{11} = 0.975$$

Decisions: a record at j wins by stopping with $\frac{j}{30}$ and by continuing with $\frac{j}{30}\sum_{i=j}^{29}\frac{1}{i}$:

$$j = 11:\ 0.367 \text{ vs } 0.379 \to \text{continue}$$

$$j = 12:\ 0.400 \text{ vs } 0.377 \to \textbf{hire}$$

A 3×10^5-run simulation confirms $\mathbb{P} = 0.380$, $\mathbb{E}[T] = 22.38$.

⋆ *Full-circle intuition:* one comparison — "chance this record is the best versus the chance a later record beats it" — gives the cut-off, success rate, stopping law, and the near-$\frac{1}{e}$ behaviour already at $n = 30$.